Digital Signal Processing in VLSI Design

Homework 4
CORDIC

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May 2026

Contents

1 CORDIC Scaling Factor Analysis (Step 1)

The CORDIC algorithm introduces a scaling factor at each micro-rotation step i:

$$K_i = \frac{1}{\sqrt{1+2^{-2i}}}, \quad i = 0, 1, 2, \ldots$$

After N micro-rotations the cumulative scaling factor is

$$S(N) = \prod_{i=0}^{N-1} K_i = \prod_{i=0}^{N-1} \frac{1}{\sqrt{1+2^{-2i}}}.$$

Figure 1 shows $S(N)$ as a function of N for $N = 1, \ldots, 30$.

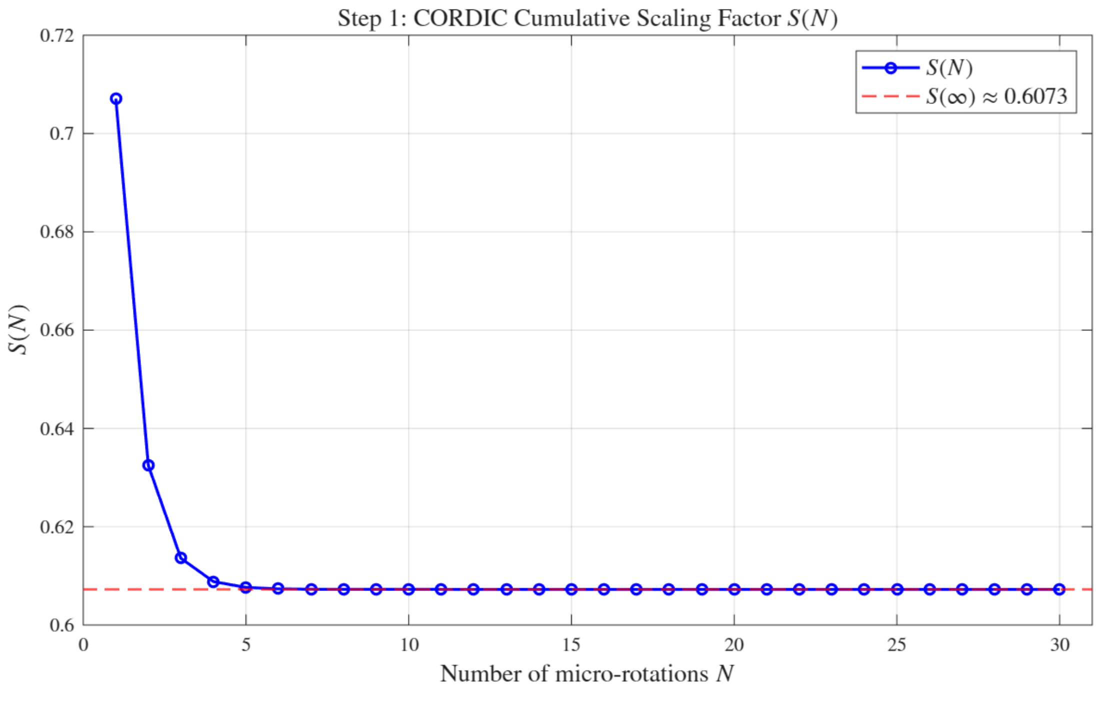

Figure 1: $S(N)$ vs. N.

Result: $S(N)$ converges to **0.607253** (the reciprocal of the CORDIC gain constant $K \approx 1.6468$). Beyond $N \approx 10$ the change per step is negligible, so $N = 10$ is sufficient for double-precision accuracy.

2 Fixed-Point Word-Length Selection (Step 2)

2.1 Integer Bit Requirement

During CORDIC rotations the vector (X, Y) is not normalized; its magnitude grows by $1/S(N)$. For unit-circle inputs ($|X^2 + Y^2| = 1$) the maximum intermediate value is

$$\max |X(i)| \leq \frac{1}{S(\infty)} \approx 1.6468.$$

One integer bit provides the range $[-2, 2)$, which covers this growth. Therefore the **X/Y format** is $1\mathrm{S} + 1\mathrm{I} + w\mathrm{F}$ ($w + 2$ bits total).

2.2 Fractional Bit Sweep

With $S = 30$ micro-rotations and floating-point angles (isolating the X/Y word-length effect), the average absolute phase error over the ten test angles $\alpha_m = \frac{4m+2}{20}\pi$, $m = 0, \ldots, 9$, is swept for $w = 1, \ldots, 16$.

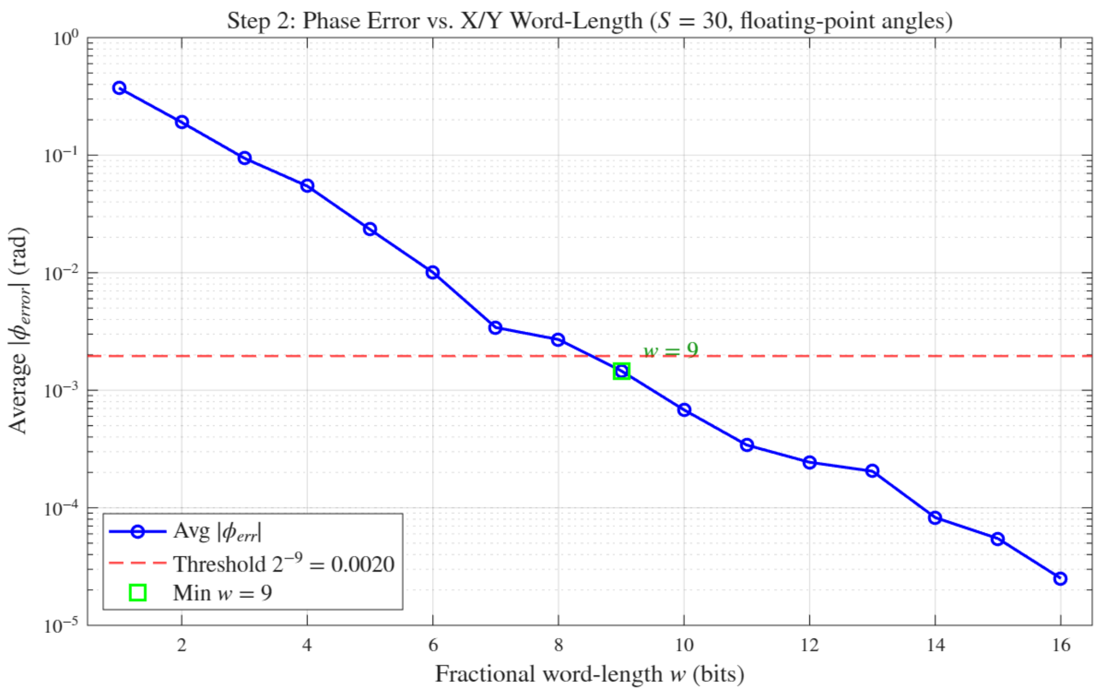

Figure 2: Average phase error vs. X/Y fractional word-length w. The dashed line marks the threshold 2^{-9}.

Result: The minimum w satisfying average $|\phi_{\mathrm{err}}| < 2^{-9}$ is $\mathbf{w = 9}$. The total X/Y word-length is therefore $\mathbf{W = 11}$ bits ($1\mathrm{S} + 1\mathrm{I} + 9\mathrm{F}$, scale $2^9 = 512$). The hardware implementation uses $w = 12$ ($W = 14$ bits) to accommodate additional quantization effects not captured in this sweep; see Section 3.3 for details.

3 Rotation Count and Angle Word-Length (Step 3)

3.1 Part A: Sweep Over Number of Micro-Rotations N

Inputs are pre-quantized with $w = 9$. Angles are kept in floating-point to isolate the effect of N.

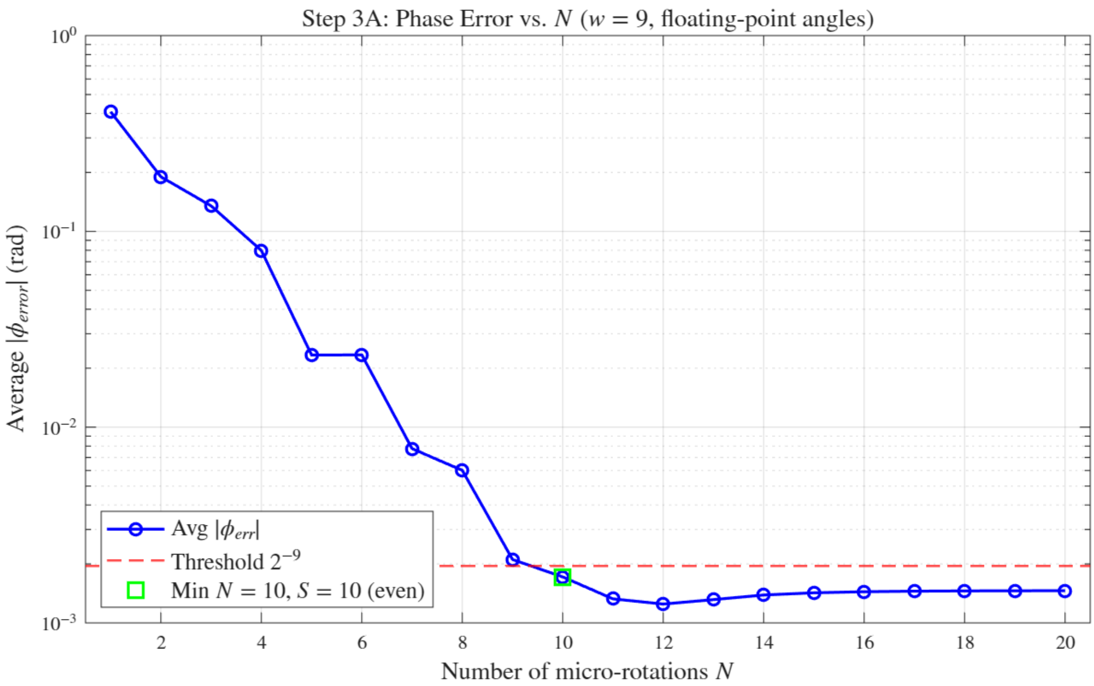

Figure 3: Average phase error vs. N (floating-point angles, $w = 9$).

Result: The minimum N meeting the threshold is $\mathbf{N_{\min} = 10}$. Since $N_{\min}$ is already even, the Step-7 S/2-unfolding architecture uses: $\mathbf{S = 10}$. The hardware uses $S = 12$ for tighter margins; see Section 3.3.

3.2 Part B: Sweep Over Angle Word-Length a_w

With $S = 10$ fixed, a_w is swept from 4 to 20.

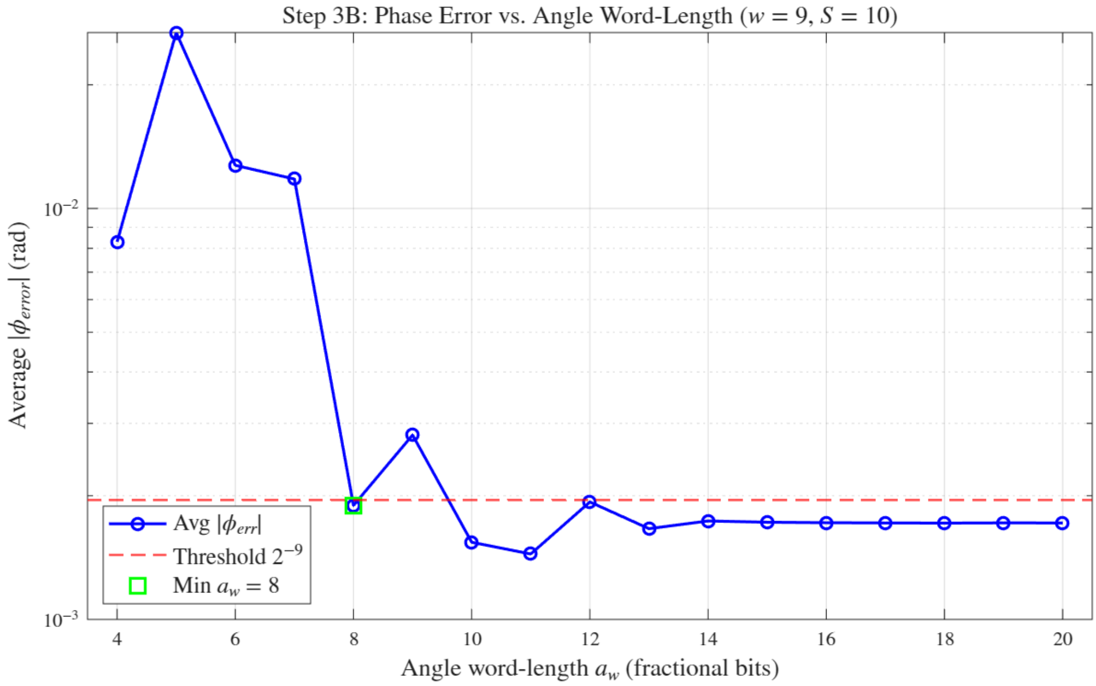

Figure 4: Average phase error vs. angle word-length a_w ($w = 9$, $S = 10$).

Result: The minimum a_w satisfying the threshold is $\mathbf{a_w = 8}$. The **theta accumulator format** is therefore 1S + 2I + 8F, total $\mathbf{TW = 11}$ bits (scale $2^8 = 256$; range $[-\pi, \pi]$ requires 2 integer bits since $\pi \approx 3.14$). The hardware uses $a_w = 10$ ($TW = 13$ bits); see Section 3.3.

3.3 Elementary Angle LUT

Table 1 lists the hardware LUT entries $\theta_e(i) = \arctan(2^{-i})$ quantized to $a_w = 10$ unsigned fractional bits (hardware configuration).

Table 1: Elementary angle LUT ($a_w = 10$, unsigned, scale $2^{10} = 1024$).

i	Float (rad)	Integer	Binary
0	0.785398	804	0.1100100100
1	0.463648	475	0.0111011011
2	0.244979	251	0.0011111011
3	0.124355	127	0.0001111111
4	0.062419	64	0.0001000000
5	0.031240	32	0.0000100000
6	0.015624	16	0.0000010000
7	0.007812	8	0.0000001000
8	0.003906	4	0.0000000100
9	0.001953	2	0.0000000010
10	0.000977	1	0.0000000001
11	0.000488	0	0.0000000000

3.4 Hardware Parameter Revision

The sweep analysis in Sections 2 and 3 assumes that only X/Y quantization and angle LUT quantization contribute to phase error. In practice, three additional quantization sources were not accounted for:

1. **Accumulated angle** θ: the theta register accumulates rounding errors from S successive LUT additions.
2. **Initial quadrant offset** ($\pm\pi$): π is itself a quantized constant; its error propagates directly into the final angle.
3. **Magnitude output**: the CSD scaling introduces a separate quantization error on top of the CORDIC gain approximation.

Because these effects were not captured, the minimum parameters from the sweep (Table 2) do not satisfy the error thresholds in hardware. Python simulation of the full fixed-point model revealed that the parameters in Table 3 are the minimum that achieve both thresholds simultaneously.

Table 2: Minimum parameters from sweep analysis (Table 6.1 — analytical lower bound).

Parameter	**Format**	**Description**
Input X and Y	1S-1I-9F	11-bit, scale $2^9 = 512$
Elementary Angle θ_e	1S-2I-8F	11-bit, scale $2^8 = 256$
Accumulated Angle θ	1S-2I-8F	11-bit (not error-analyzed)
Initial Rotation Angle θ_i	1S-2I-8F	11-bit (not error-analyzed)
Micro-rotations S	—	10 iterations
Magnitude Output	1S-1I-9F	11-bit (not error-analyzed)

Table 3: Actual hardware parameters satisfying all error constraints (Table 6.2).

Parameter	**Format**	**Description**
Input X and Y	1S-1I-12F	14-bit, scale $2^{12} = 4096$
Elementary Angle θ_e	1S-2I-10F	13-bit, scale $2^{10} = 1024$
Accumulated Angle θ	1S-2I-10F	13-bit, matches θ_e
Initial Rotation Angle θ_i	1S-2I-10F	13-bit, matches θ_e and θ
Micro-rotations S	—	12 iterations
Magnitude Output	1S-1I-12F	14-bit, matches input format

4 Number of Micro-Rotations for Magnitude (Step 4)

The residual rotation angle after N CORDIC steps introduces a magnitude error bounded by

$$\epsilon(N) \le 1 - \cos(\theta_e(N-1)) = 1 - \frac{1}{\sqrt{1 + 2^{-2(N-1)}}}.$$

Figure 5 shows this bound vs. N.

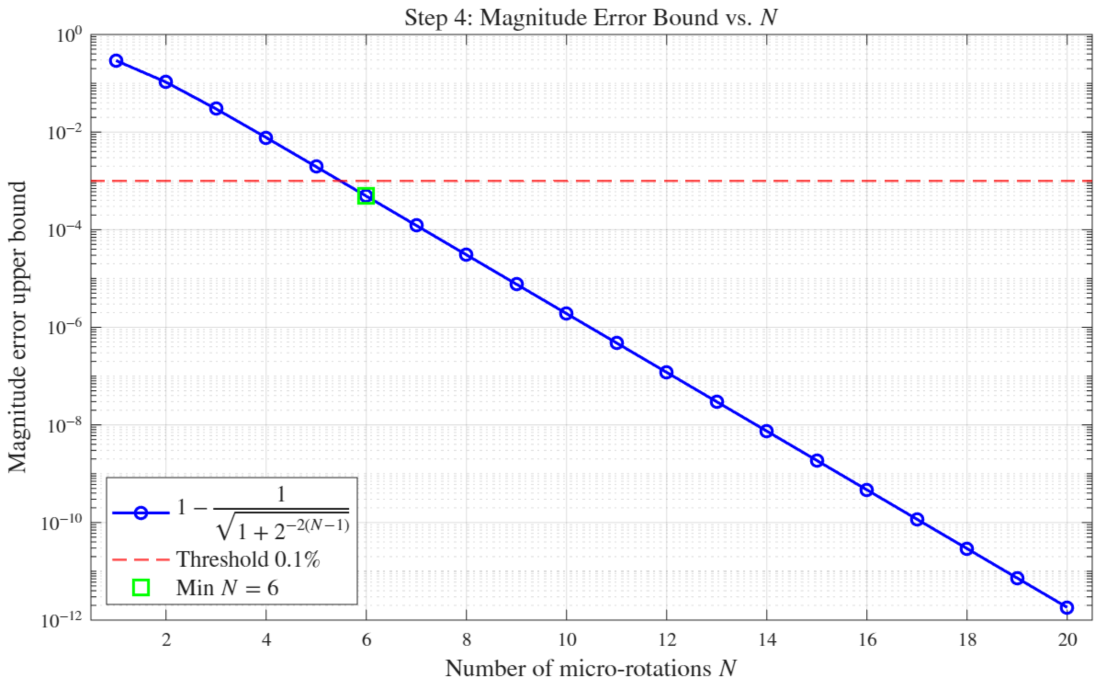

Figure 5: Analytical magnitude error bound $\epsilon(N)$ vs. N. The dashed line marks the 0.1% threshold.

Result: The minimum N satisfying $\epsilon(N) < 0.1\%$ is $\mathbf{N}_{\text{mag}} = \mathbf{6}$. Since $S = 12 \ge 6$, the hardware-chosen $S = 12$ is sufficient for both arctangent and magnitude computation.

5 CSD Representation of Scaling Factor (Step 5)

5.1 Motivation

After $N = 12$ CORDIC rotations, $X(N) = |Z|/S(12) \approx |Z|/0.60725$. To recover the true magnitude we multiply by $S(12)$. Using Canonical Signed Digit (CSD) representation, this multiplication reduces to a **shift-and-add** network with no multiplier.

5.2 CSD Approximation Error vs. Word-Length

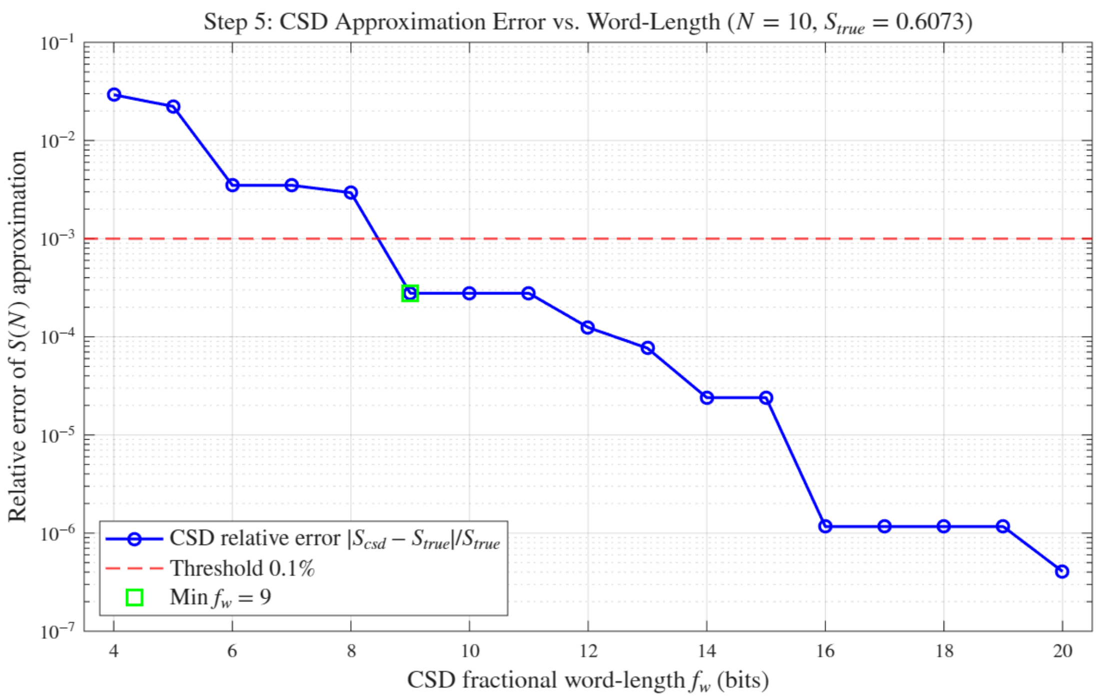

Figure 6: CSD relative approximation error vs. fractional word-length f_w. The dashed line marks the 0.1% threshold.

Result: The minimum CSD word-length satisfying $|S_{\text{CSD}} - S_{\text{true}}|/S_{\text{true}} < 0.1\%$ is $\mathbf{f_w = 9}$. The chosen approximation is

$$A_N = 2^{-1} + 2^{-3} - 2^{-6} - 2^{-9} = 0.607422,$$

with a relative error of **0.028**%. This requires **3 adders** (4 non-zero CSD digits minus 1).

5.3 Shift-and-Add Implementation

The magnitude computation in hardware is:

$$\texttt{outMag} = X_s \gg 1 \; + \; X_s \gg 3 \; - \; X_s \gg 6 \; - \; X_s \gg 9,$$

where X_s denotes the final X register value after $S = 12$ micro-rotations.

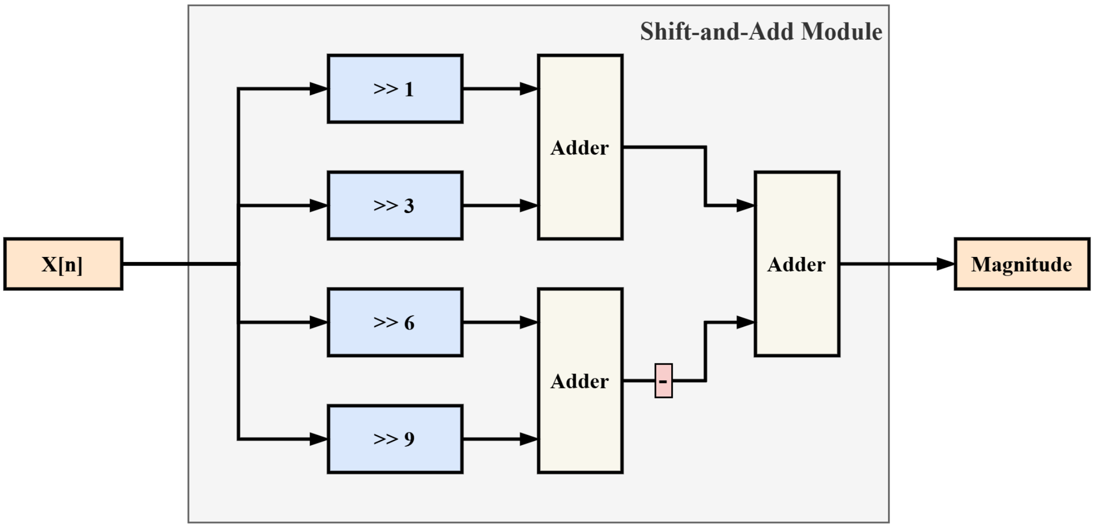

Figure 7: CSD shift-and-add module.

6 Iterative CORDIC Architecture (Step 6)

6.1 Architecture Overview

Module `CORDIC.v` implements the iterative vectoring-mode CORDIC. The interface is:

$$\text{clk, rst\_n,} \quad \texttt{inX}[13:0],\ \texttt{inY}[13:0],\ \texttt{in\_valid} \ \longrightarrow\ \texttt{outTheta}[12:0],\ \texttt{out\_valid}.$$

FSM: The control uses a three-state FSM: **IDLE** → **ITERATE** ($S = 12$ cycles) → **DONE**. Total latency from `in_valid` to `out_valid`: $\mathbf{S+2=14}$ clock cycles (1 input DFF + S iterations + 1 output/DONE cycle).

Initial-Stage Quadrant Mapping: CORDIC converges only within $\pm 90°$. Inputs in Q2/Q3 are mapped to Q1/Q4:

$$X < 0 \implies \begin{cases} X_0 = -X, \quad Y_0 = -Y \\ \theta_{\text{init}} = +\pi & (Y \geq 0) \\ \theta_{\text{init}} = -\pi & (Y < 0) \end{cases}$$

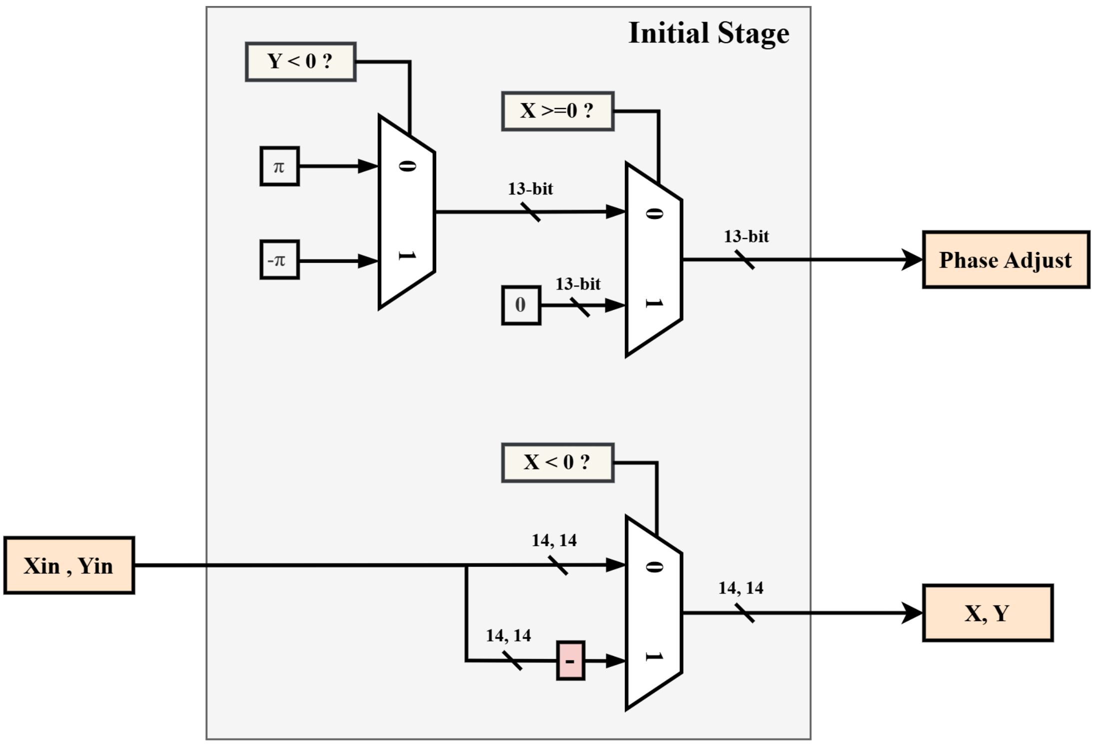

Figure 8: Initial Stage block diagram: quadrant mapping for (X, Y) and phase offset selection for θ_{init}.

Micro-Rotation Rule: At iteration i, the direction $\mu_i = -\text{sign}(Y_i)$:

$$\begin{aligned} X_{i+1} &= X_i - \mu_i \,(Y_i \gg i), \\ Y_{i+1} &= Y_i + \mu_i \,(X_i \gg i), \\ \theta_{i+1} &= \theta_i - \mu_i \,\theta_e(i). \end{aligned}$$

6.2 Behavioral Simulation Results

Four test angles $(\alpha_0, \alpha_3, \alpha_6, \alpha_9)$ covering all four quadrants are verified in `USE_ITERATIVE` mode.

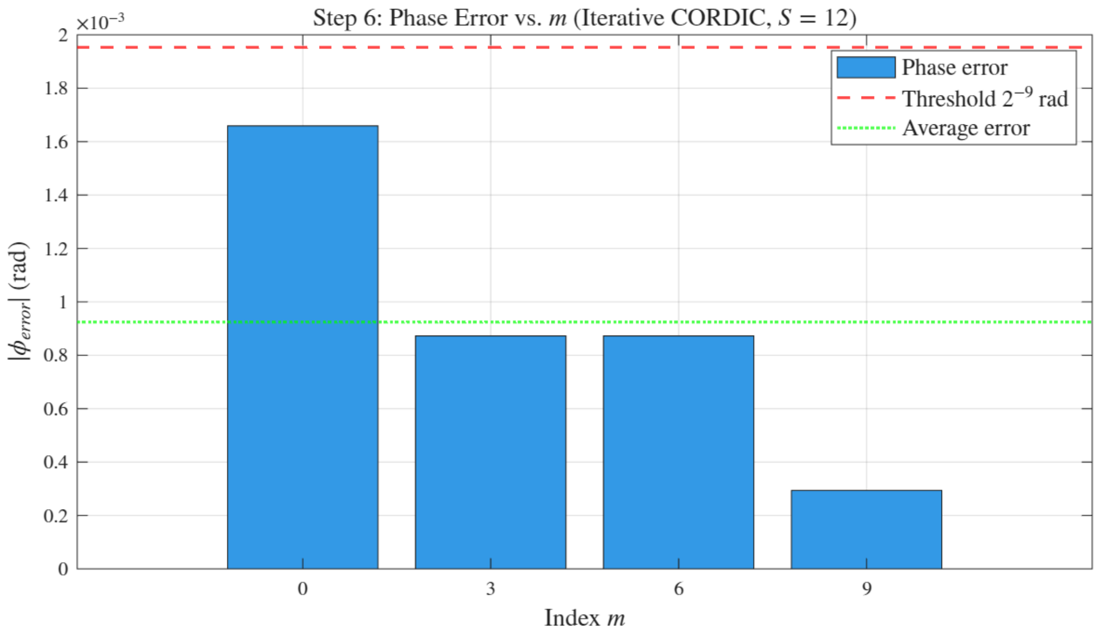

Figure 9: Step 6: Phase error (output − golden) for $m = 0, 3, 6, 9$. All mismatches = 0 confirms bit-exact match with the MATLAB golden model.

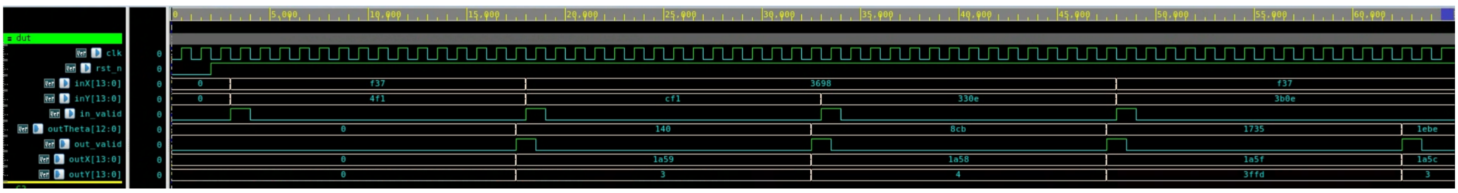

Figure 10: Step 6: Behavioral simulation waveform (4 inputs: $\alpha_0, \alpha_3, \alpha_6, \alpha_9$).

7 S/2-Unfolded CORDIC Architecture (Step 7)

7.1 Architecture Overview

Module `CORDIC_unfolded.v` replaces the iterative FSM with a two-stage pipeline, achieving a throughput of one result every 2 clock cycles (latency = 2 cycles).

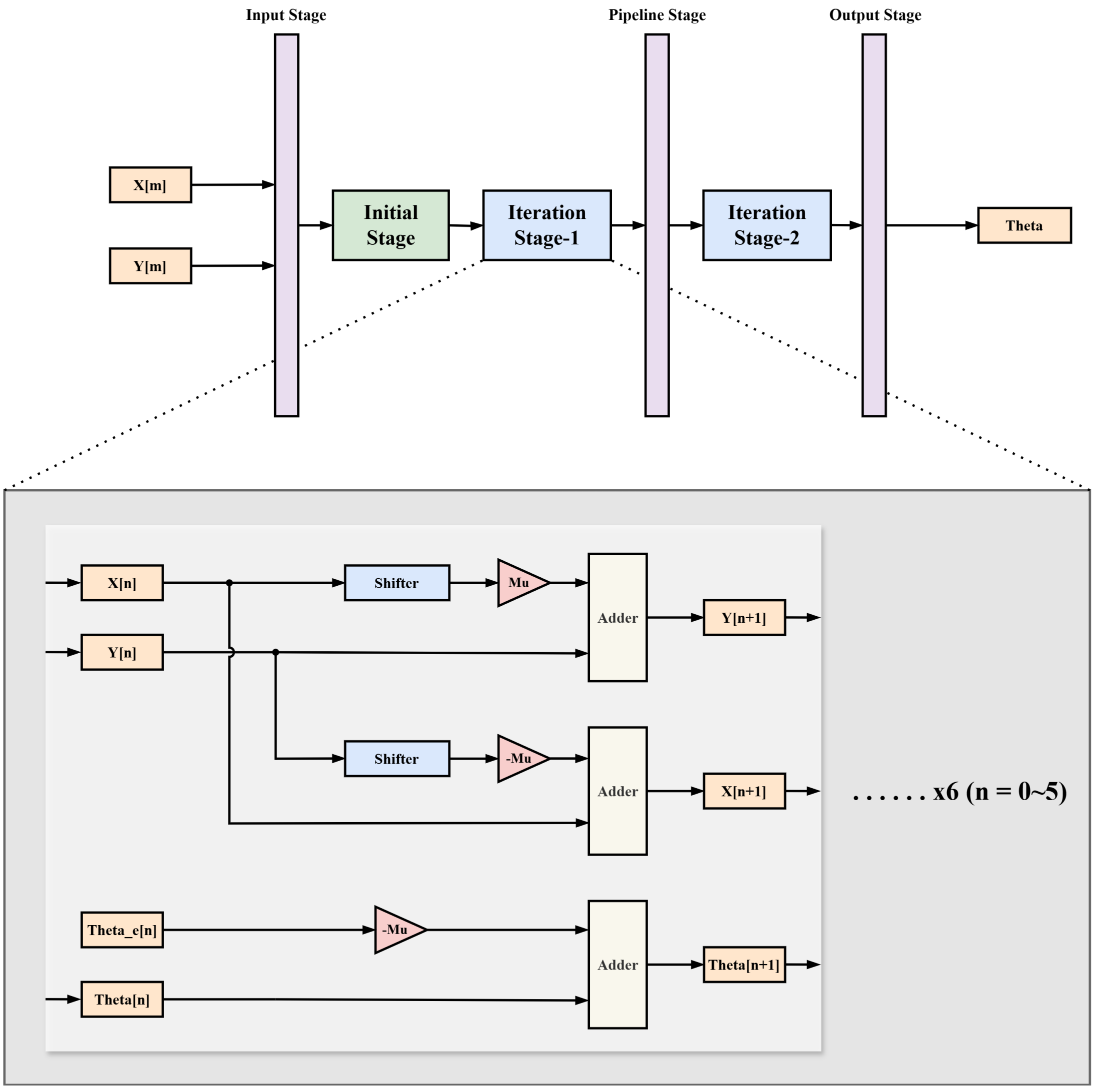

Figure 11: S/2-Unfolded CORDIC architecture.

Pipeline structure:

$$\text{Input DFF (cycle 0)} \rightarrow \text{Initial stage} + \text{Stages 0–5 (comb.)} \rightarrow \text{Pipeline FF (cycle 1)}$$
$$\rightarrow \text{Stages 6–11 (comb.)} \rightarrow \text{Output FF (cycle 2)}$$

Each stage i is a hardcoded combinational block (no barrel shifter):

$$Y[W-1] = 0 : \quad X' = X + (Y \gg i),\ Y' = Y - (X \gg i),\ T' = T + A_i,$$

$$Y[W-1] = 1 : \quad X' = X - (Y \gg i),\ Y' = Y + (X \gg i),\ T' = T - A_i.$$

7.2 Circuit Synthesis and Timing Constraints

The timing constraints applied during synthesis are summarised in Table 4.

Table 4: Timing constraints applied during synthesis (TSMC 16 nm FinFET (ADFP)).

Constraint	**Value**
Clock period T_{clk}	1.0 ns (1 GHz)
Clock uncertainty	0.5 ns
Clock source latency	0.1 ns
Input max delay	0.5 ns
Output max delay	0.5 ns

Both the S/2-unfolded design and the iterative design meet timing with no setup or hold violations under this constraint.

7.3 Behavioral Simulation Results

All ten test angles ($m = 0, \ldots, 9$) are verified.

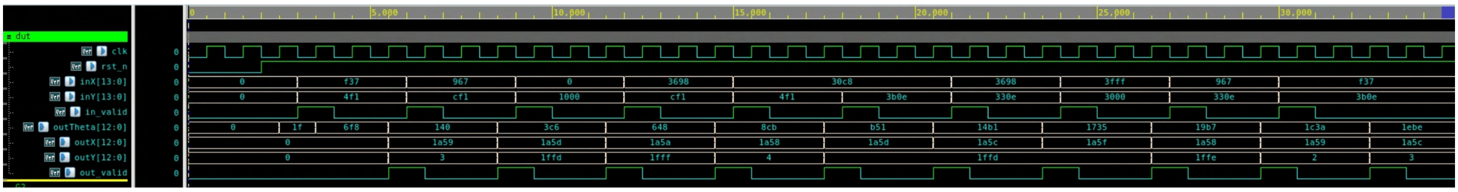

Figure 12: Step 7: Behavioral simulation waveform (10 inputs, $m = 0, \ldots, 9$).

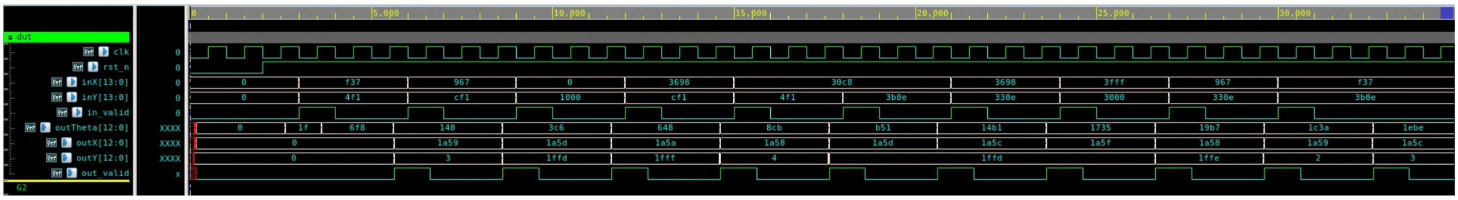

Figure 13: Step 7: Post-synthesis simulation waveform (10 inputs, with SDF back-annotation).

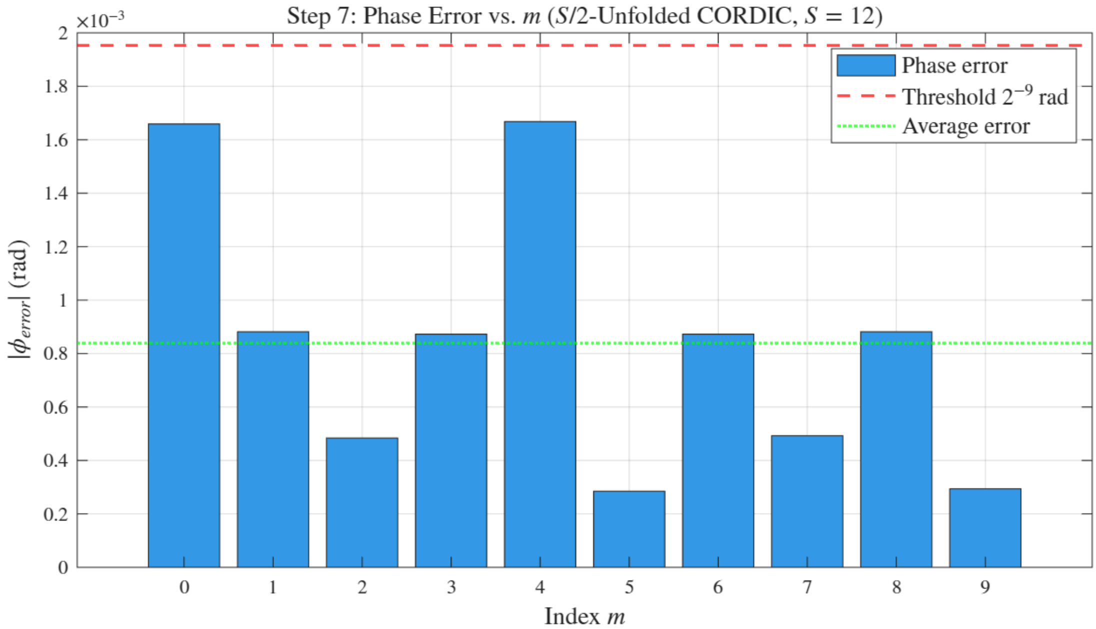

Figure 14: Step 7: Phase error between post-synthesis simulation results and floating-point arctangent reference ($m = 0, \ldots, 9$). Results are identical to behavioral simulation, confirming bit-exact agreement after synthesis.

8 Area Comparison: Iterative vs. Unfolded (Step 8)

Both `CORDIC.v` (iterative) and `CORDIC_unfolded.v` (S/2-unfolded) are synthesized under the same timing constraint ($T_{\text{clk}} = 1\,\text{ns}$, i.e. **1 GHz**) using Design Compiler with the TSMC 16 nm FinFET (ADFP) slow corner library. Both designs meet timing with no setup or hold violations.

Table 5: Post-synthesis area comparison (TSMC 16 nm FinFET (ADFP), $T_{\text{clk}} = 1\,\text{ns}$, 1 GHz).

Design	**Total Cell Area (μm^2)**	**Ratio**
`CORDIC` (iterative)	310	1.00×
`CORDIC_unfolded` (S/2-unfolded)	987	3.18×

Analysis: The S/2-unfolded design occupies 987/310 ≈ **3.18**× the area of the iterative design. This increase comes from instantiating the 12 micro-rotation stages as combinational hardware instead of reusing one iterative datapath. The trade-off is area for throughput: the iterative design requires $S + 2 = 14$ cycles per output, while the unfolded design produces one result every 2 clock cycles.

9 Full CORDIC with Magnitude Output (Step 9)

9.1 Architecture Overview

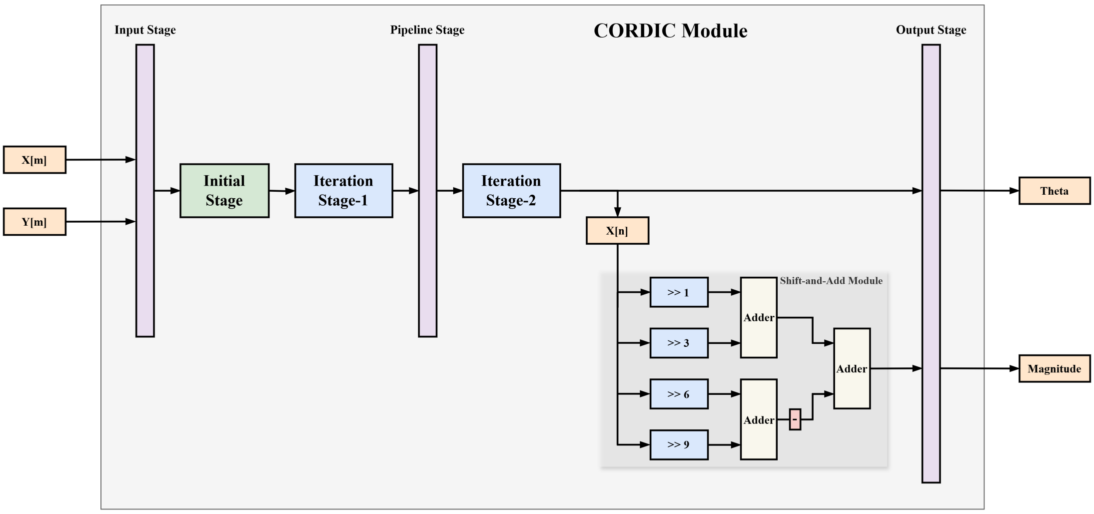

Figure 15: Full CORDIC architecture with magnitude output.

Module `CORDIC_mag.v` extends the S/2-unfolded design with a CSD shift-and-add magnitude scaler. The same timing constraints as Step 7 (Table 4) are applied, and the 2-cycle pipeline is preserved:

$$\texttt{outMag} = X_{11} \gg 1 + X_{11} \gg 3 - X_{11} \gg 6 - X_{11} \gg 9,$$

where X_{11} is the post-rotation X value, approximately $|Z| \cdot 1/S(12)$.

9.2 Behavioral Simulation Results

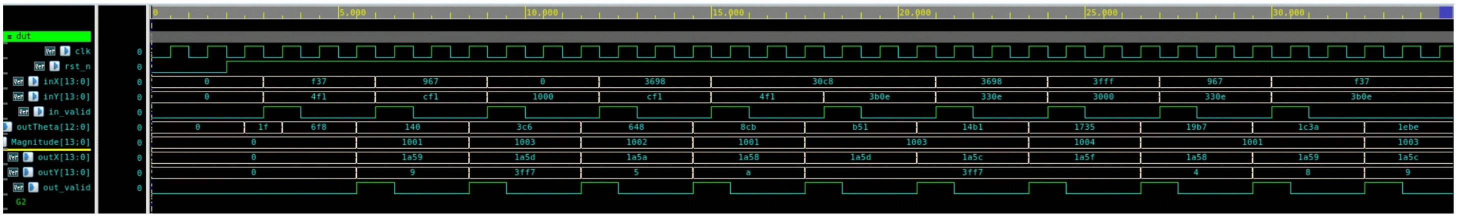

Figure 16: Step 9: Behavioral simulation waveform (10 inputs, magnitude + phase outputs).

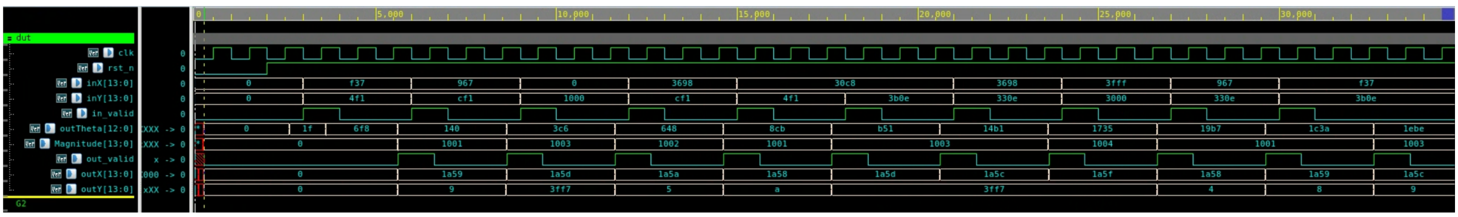

Figure 17: Step 9: Post-synthesis simulation waveform (10 inputs, with SDF back-annotation).

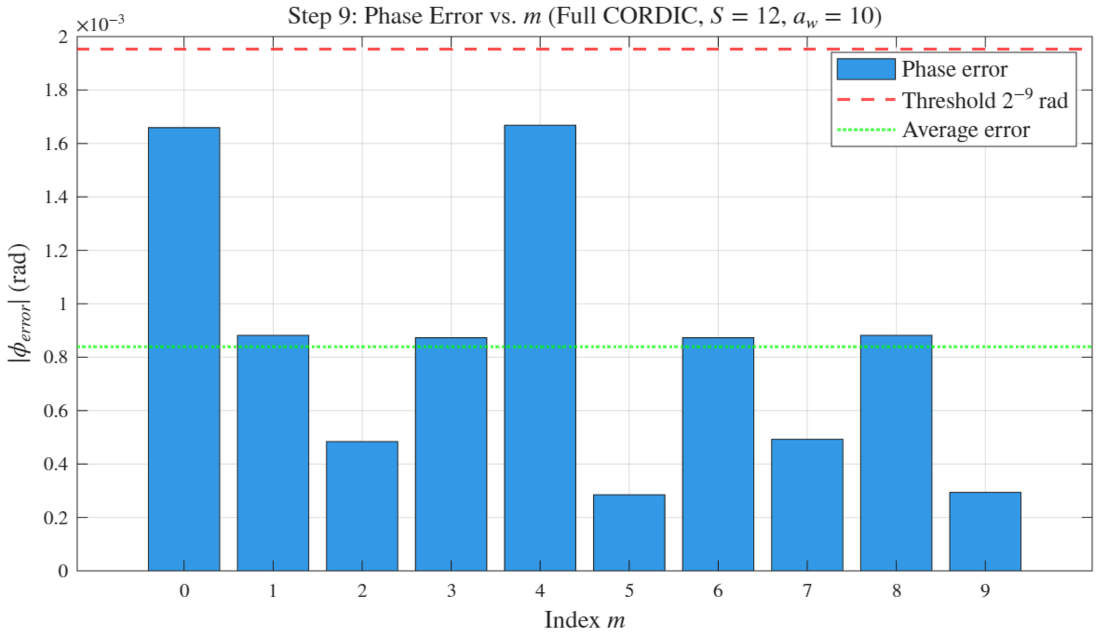

Figure 18: Step 9: Phase error between post-synthesis simulation results and floating-point arctangent reference ($m = 0, \ldots, 9$). Results are identical to behavioral simulation, confirming bit-exact agreement after synthesis.

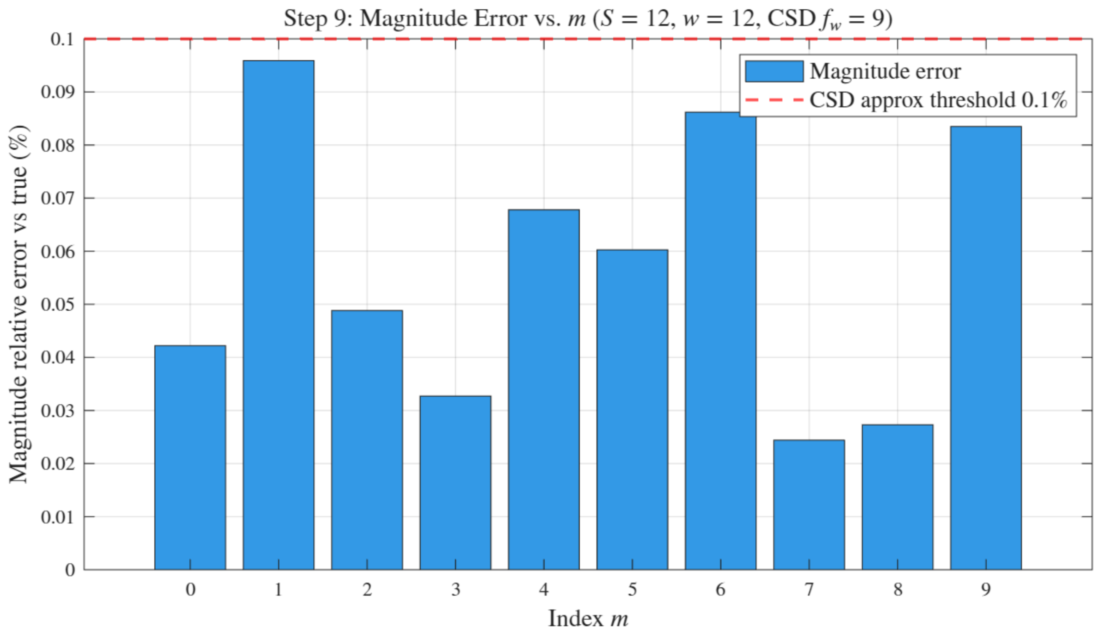

Figure 19: Step 9: Magnitude error (%) between the post-synthesis magnitude output and floating-point magnitude $\sqrt{X^2 + Y^2}$ ($m = 0, \ldots, 9$). Maximum error $< 0.1\%$, consistent with the CSD approximation bound.